

Demonstration of Proposed Features

Date	2 July 2026
Project Name	OptiCrop
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase for the OptiCrop project and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	Smart Crop Recommendation	Accepts N, P, K, temperature, humidity, pH and rainfall inputs from the farmer and recommends the most suitable crop for maximum yield using a trained ML model.	Implemented	Yes	Validated with sample soil datasets; recommendation accuracy verified.
2	Crop Suitability & Environmental Assessment	Evaluates whether current soil and climate conditions are suitable for a specific crop chosen by the user and returns a compatibility score.	Implemented	Yes	Suitability score and productivity potential displayed to user.
3	Data Input & Validation Module	Provides a form-based interface for entering N, P, K, temperature, humidity, pH and rainfall values, with range checks to ensure valid agronomic inputs.	Implemented	Yes	Input boundaries cross-checked against dataset ranges.
4	Research & Policy Analytics Dashboard	Visual dashboard for researchers and policymakers to analyze crop-environment relationships, trends and patterns across regions for policy planning.	Partial	Yes	Core charts implemented; regional comparison filters planned for next iteration.
5	Yield & Resource Optimization Suggestions	Generates actionable recommendations on resource usage (fertilizer, irrigation) to improve yield and promote sustainable farming practices.	Partial	No	Logic implemented; UI integration pending for live demo.

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6	Result Visualization & Report Export	Displays recommendation results with clear visual indicators and allows farmers/researchers to export a summary report of the analysis.	Implemented	Yes	Exportable report tested with sample scenarios.

Feature Implementation Summary:

Total Features Proposed	6
Total Features Implemented	4
Total Features Demonstrated	5
Overall Implementation Rate (%)	67%